

Richard Álvarez

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Professional Summary

Applied AI systems engineer building production agent systems, voice workflows, and enterprise integrations. Experienced in authorization, human approval controls, automated testing, observability, and reliable deployment across TypeScript and Python stacks.

Experience

AI Solutions Engineer Sep 2025 – Present
Layer 7 Systems | Independent Contractor Remote

- Built and operated a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph and Teams for structured service desk intake, ticket actions, and technician routing.
- Engineered caller and company resolution, contact-safe ticket access, backend authorization gates, deterministic transfer eligibility, post-call workflows, and human fallback to prevent unsafe writes and cross-client data exposure.
- Developed an authenticated React operations dashboard and expanded and maintained a 750+ test regression suite covering authorization, validation, transfers, external API behavior, alerts, and failure recovery.
- Deployed Docker workloads through GitHub Actions across staging and production, then diagnosed duplicate tickets, stale identity state, API drift, retry and dead-letter queue failures, transfer defects, and CI and VPS incidents.
- Built a workflow-analysis pipeline that reviewed 60 recordings from 14 users and identified 176 AI-assistable moments, then delivered a Next.js and Neon training platform with four courses and 16 modules for 34 trainees.

Freelance Web Developer Oct 2024 – Present
Richard Álvarez | Sole Proprietorship Chicago, IL

- Design and ship responsive web applications for creative and commercial clients using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, and object storage.

IT Assistant Sep 2023 – Feb 2024
Kenyon College Library & Information Services Gambier, OH

- Supported workstation imaging, hardware deployment, printer and AV troubleshooting, and secure drive destruction, and built a spreadsheet tool to automate classroom maintenance scheduling.

Selected Project

Vigil | TypeScript, Bun, Hono, SQLite FTS5, Next.js, Cloudflare | GitHub 2026

- Built an open-source, self-hosted email intelligence agent with multi-model routing, encrypted user-supplied credentials, persistent memory, deterministic action mapping, alerts, audit logs, and per-invocation cost tracking.

Education

Kenyon College Aug 2020 – May 2024
Bachelor of Arts in Film Gambier, OH
Minor in History; Integrated Program in Humane Studies concentration

Courses: Software Development, Data Structures and Programming Design, AI for the Humanities

Technical Skills

Agent Systems: MCP, tool calling, RAG, structured outputs, context and memory design, agent orchestration
Workflow and Integration: n8n, REST APIs, webhooks, Autotask, Microsoft Graph and Teams, identity and permissions, human approval controls
Reliability and Delivery: Docker, GitHub Actions, Linux and VPS, CI/CD, automated testing, validation, logging, observability, incident response, cost monitoring
Product and Data: TypeScript, Python, Bun and Node.js, Hono and FastAPI, React and Next.js, PostgreSQL, SQLite FTS5
Systems and Media: C++20, CMake, FFmpeg, PySceneDetect